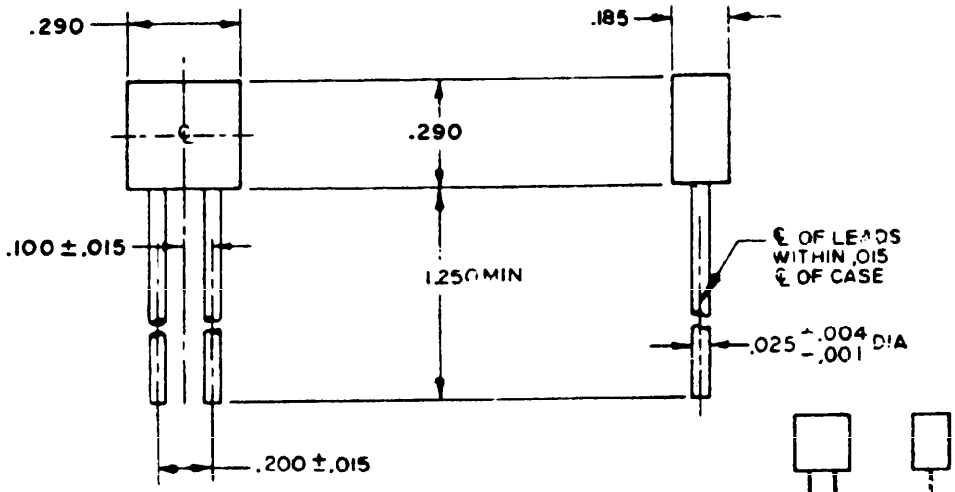
MILITARY SPECIFICATION SHEET

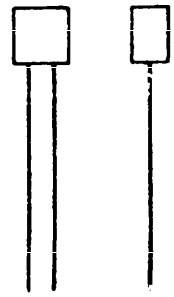
CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), ESTABLISHED RELIABILITY, SYTLE CKR07

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-39014.



INCHES	MM	INCHES	MM
.001	.03	.100	2.54
.004	.10	.185	4.70
.010	.25	.200	5.08
.015	.38	.290	7.37
.025	.64	1.250	31.75



APPROX ACTUAL SIZE

NOTES:
1. Dimenions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010.

FIGURE 1. Dimensions and configuration.

Ⓓ denotes changes

FSC 5910

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REQUIREMENTS:

Dimensions and configuration: See figure.

Case type: Molded or preformed.

Capacitance value: See table I.

Capacitance tolerance: 10 percent.

Operating temperature range: -55°C to +125°C.

Voltage rating: 50 volts dc.

Dissipation factor: Not more than 2.5 percent.

Dielectric withstanding voltage:

Body insulation: Shall be able to withstand 1,300 volts between leads and case.

Insulation resistance:

At 25°C - Not less than 100,000 megohms or 1,000 megohm-microfarads whichever is less.

At 125°C - Not less than 10,000 megohms or 100 megohm-microfarads whichever is less.

Insertion loss: Not applicable.

Seal: Not applicable.

Voltage-temperature limits: In accordance with MIL-C-39014.

Capacitance change with reference to 25°C	
Steps A through D of table V Bias = 0 volts	Steps E through G of table V Bias = rated voltage
±15 percent	+15, -25 percent

Immersion:

IR - Not less than 50 percent of initial 25°C requirement.

Salt spray (corrosion): Not applicable.

Resistance to soldering heat:

ΔC - +20, -0 percent from initial measured value.

Life:

Ⓓ Rated conditions, 100 percent of dc rated voltage at 125°C.
0 through 3,000 hours:
IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC - ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.

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4,000 through 32,000 hours:
IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC - ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.

Accelerated conditions, 200 percent of dc rated voltage at 125°C.
0 and 250 hours:
IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC - ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.

(D) 1,000, 2,000, and 4,000 hours:
IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC - ±20 percent from initial measured value.
Dissipation factor - Not greater than 2.5 percent.

Part number: M39014/03- (dash number from table I).

Supersession data: See table II.

D TABLE I. Dash number and electrical characteristics.

Dash number				Capacitance
Failure rate level (%/1,000 hours)				
1.0 (M)	0.1 (P)	0.01 (R)	0.001 (S)	
				pF
0306	0311	0316	0321	470,000
0307	0312	0317	0322	560,000
0308	0313	0318	0323	680,000
0309	0314	0319	0324	820,000
0310	0315	0320	0325	1,000,000

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Ⓓ TABLE II. Substitutability data.

Dash no. in MIL-C-39014/3D 1/				Dash no. in MIL-C-39014/3C 1/					Substitutable for dash no. in:														
									MIL-C-39014/3B and /1B					MIL-C-39014/3A					MIL-C-39014/3				
M	P	R	S	L	M	P	R	S	L	M	P	R	S	L	M	P	R	S	L	M	P	R	S
0306	0311	0316	0321	0301	0306	0311	0316	0321	0201	0206	0211	0216	0221	0101	0106	0111	0116	0121	0001	0006	0011	0016	0021
0307	0312	0317	0322	0302	0307	0312	0317	0322	0202	0207	0212	0217	0222	0102	0107	0112	0117	0122	0002	0007	0012	0017	0022
0308	0313	0318	0323	0303	0308	0313	0318	0323	0203	0208	0213	0218	0223	0103	0108	0113	0118	0123	0003	0008	0013	0018	0023
0309	0314	0319	0324	0304	0309	0314	0319	0324	0204	0209	0214	0219	0224	0104	0109	0114	0119	0124	0004	0009	0014	0019	0024
0310	0315	0320	0325	0305	0310	0315	0320	0325	0205	0210	0215	0220	0225	0105	0110	0115	0120	0125	0005	0010	0015	0020	0025

1/ Dash numbers in lower FR level columns are substitutable for dash numbers in all higher FR level columns.

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Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AR, MI
Navy - TD
Air Force - 17, 85, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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